

IGARKA, Russian Federation

WMO: 232740

Lat: 67.4423N

Lon: 86.6053E

Elev: 28

StdP: 100.99

Time Zone: 7.00 (E07)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-46.7	-44.7	-49.1	0.0	-46.6	-47.2	0.0	-44.5	9.6	-15.9	8.9	-18.0	0.6	330	0.791

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	26.4	17.8	24.2	17.0	21.8	15.9	19.2	24.2	17.9	22.2	16.6	20.9	3.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.3	12.4	21.5	16.0	11.4	20.6	14.7	10.5	18.8	54.9	24.3	50.7	22.2	46.8	21.1	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.6	6.9	DB	-48.0	29.4	3.5	2.2	-50.5	30.9	-52.6	32.2	-54.6	33.4	-57.1	35.0	(4)
(5)			WB	-48.1	20.9	3.3	1.5	-50.5	22.0	-52.4	22.9	-54.3	23.7	-56.7	24.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-7.0	-27.7	-25.7	-16.9	-9.2	-0.7	12.0	15.7	11.7	5.5	-5.2	-20.5	-24.4	(6)	
	DBStd	17.02	10.29	8.90	9.16	7.73	5.67	5.38	4.15	3.77	4.29	6.63	10.12	10.23	(7)	
	HDD10.0	6578	1169	1000	834	576	334	41	4	23	146	471	913	1068	(8)	
	HDD18.3	9281	1427	1233	1093	826	589	199	104	207	384	729	1163	1326	(9)	
	CDD10.0	373	0	0	0	0	3	101	179	78	12	0	0	0	(10)	
	CDD18.3	32	0	0	0	0	0	8	22	2	0	0	0	0	(11)	
	CDH23.3	289	0	0	0	0	2	79	194	13	0	0	0	0	(12)	
	CDH26.7	53	0	0	0	0	0	11	42	1	0	0	0	0	(13)	
(14) Wind	WSAvg	3.2	3.2	3.0	3.2	3.6	3.5	3.3	3.0	2.9	3.2	3.4	2.9	3.3	(14)	
(15) Precipitation	PrecAvg	494	30	24	27	28	27	47	53	61	56	58	42	37	(15)	
	PrecMax	641	63	49	48	77	56	108	143	117	132	99	74	63	(16)	
	PrecMin	360	6	3	5	2	3	6	11	17	19	21	5	15	(17)	
	PrecStd	78	14	10	11	15	12	22	32	25	24	17	16	11	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.1	-4.2	1.8	9.2	19.8	28.2	29.9	25.2	20.2	9.8	1.0	-3.1	(19)	
		MCWB	-6.5	-4.5	-0.2	4.2	12.8	17.1	20.4	18.8	14.7	8.6	0.5	-3.6	(20)	
	2%	DB	-8.8	-6.9	-0.3	6.0	13.0	25.7	27.7	22.3	16.2	6.1	-0.7	-4.9	(21)	
		MCWB	-9.3	-7.3	-1.4	2.5	7.8	16.0	19.0	16.8	12.3	5.3	-1.1	-5.2	(22)	
	5%	DB	-11.8	-9.7	-2.0	3.6	9.6	23.1	25.5	20.0	13.8	4.0	-3.0	-7.6	(23)	
		MCWB	-12.1	-10.1	-3.0	1.3	5.5	15.6	17.9	15.6	11.4	3.2	-3.5	-8.0	(24)	
	10%	DB	-14.0	-12.9	-4.1	1.8	6.9	20.7	23.0	18.0	12.0	2.2	-5.8	-10.4	(25)	
		MCWB	-14.4	-13.3	-5.0	0.3	3.8	14.5	17.0	14.6	10.0	1.5	-6.2	-10.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.5	-4.6	0.1	4.9	12.6	19.1	21.5	20.2	15.2	8.6	0.5	-3.5	(27)	
		MCDB	-6.1	-4.3	1.2	8.2	18.8	25.2	28.1	23.6	18.9	10.0	0.7	-3.1	(28)	
	2%	WB	-9.4	-7.4	-1.4	3.0	8.1	17.4	20.0	17.7	13.3	5.4	-1.2	-5.3	(29)	
		MCDB	-8.9	-7.0	-0.4	5.4	12.1	23.2	25.5	20.7	15.6	6.1	-0.8	-5.0	(30)	
	5%	WB	-12.1	-10.2	-3.0	1.6	5.7	16.2	18.7	16.1	11.7	3.1	-3.5	-8.0	(31)	
		MCDB	-11.8	-9.7	-2.0	3.2	9.2	21.4	23.6	19.5	13.5	3.7	-3.1	-7.6	(32)	
	10%	WB	-14.5	-13.3	-5.0	0.3	4.0	15.1	17.6	14.8	10.0	1.5	-6.2	-10.7	(33)	
		MCDB	-14.1	-13.0	-4.2	1.8	6.8	20.0	21.9	17.7	11.6	2.1	-5.7	-10.4	(34)	
(35) Mean Daily Temperature Range	MDBR		7.2	8.2	10.3	10.6	7.5	8.6	8.4	7.1	5.7	5.0	7.0	7.0	(35)	
	5% DB	MCDBR	8.0	9.5	8.0	8.2	9.6	12.2	12.2	10.0	9.1	4.8	6.1	7.7	(36)	
		MCWBR	8.0	9.4	7.5	6.2	6.1	5.8	5.8	5.7	6.0	4.1	6.2	7.5	(37)	
	5% WB	MCDBR	8.1	9.7	8.0	7.6	9.2	10.7	10.5	8.5	7.7	4.4	6.2	7.5	(38)	
		MCWBR	8.0	9.7	7.6	6.1	6.2	5.8	5.9	5.4	5.5	4.0	6.2	7.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.220	0.263	0.311	0.333	0.359	0.377	0.410	0.396	0.335	0.268	0.203	N/A	(40)	
	taud		1.857	1.979	2.001	2.050	2.129	2.378	2.282	2.285	2.406	2.137	1.886	N/A	(41)	
	Ebn at Noon		96	518	699	803	825	826	776	741	708	520	99	0	(42)	
	Edh at Noon		13	56	97	122	126	102	110	98	67	46	10	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.04	0.55	2.05	4.10	5.36	5.42	5.10	3.16	1.70	0.61	0.11	0.00	(44)	
	RadStd		0.01	0.06	0.23	0.39	0.48	0.46	0.63	0.37	0.20	0.11	0.02	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	+0.98	N/A	N/A	N/A	N/A	N/A	-372	-455	N/A	N/A

Nomenclature: See separate page